

- Week 5 : Data Analytics
- Lecturer : Prof. Youness Moukafih
- Lab Instructor: Prof. Safaa Berkani

Tutorial 2 – Probability

Exercise 1: Each row in the table below is a proposed grade distribution for a class. Identify each as a valid or invalid probability distribution, and explain your reasoning.

	Grades				
	A	B	C	D	F
(a)	0.3	0.3	0.3	0.2	0.1
(b)	0	0	1	0	0
(c)	0.3	0.3	0.3	0	0
(d)	0.3	0.5	0.2	0.1	-0.1
(e)	0.2	0.4	0.2	0.1	0.1
(f)	0	-0.1	1.1	0	0

Exercise 2: Suppose a survey of 1570 university students over a 4-year period was conducted. The following results were obtained regarding their faculty and the number of courses they passed:

Faculty	0-2 Courses	3-5 Courses	6+ Courses
Engineering	120	150	180
Business	200	140	60
Science	100	130	170
Arts	180	90	50

Suppose we randomly select a student from the university.

- What is the probability that the student is from the Science faculty, given they passed 3-5 courses?
- What is the probability that the student passed 0-2 courses, given they are in the Business faculty?
- What is the probability that the student passed 6+ courses, given they are in either the Engineering or Science faculties?
- What is the probability that the student is in the Arts faculty?

Exercise 3: A high school club has assessed its students and found that 65% of the students are active participants in club activities (A). Furthermore, it was found that 75% of the active participants have attended club workshops (W), while 30% of the non-active participants have not attended any workshops (W').

If a student has attended a club workshop, what is the probability that the student is not an active participant in the club activities?

Exercise 4: We would like to design a fair game for a school sports day. The game involves rolling three dice. If all three dice show the same number, you win **N** points. Otherwise, you lose one point.

- Find the value of **N** that will make the game fair.
- Using the result from i, calculate the standard deviation of the points (win/loss) for the game.

Exercise 5: Joint and conditional probabilities: $P(A) = 0.3$, $P(B) = 0.7$

- Can you compute $P(A \text{ and } B)$ if you only know $P(A)$ and $P(B)$?
- Assuming that events A and B arise from independent random processes,
 - what is $P(A \text{ and } B)$?
 - what is $P(A \text{ or } B)$?
 - what is $P(A|B)$?
- If we are given that $P(A \text{ and } B) = 0.1$, are the random variables giving rise to events A and B independent?
- If we are given that $P(A \text{ and } B) = 0.1$, what is $P(A|B)$?

Exercise 6: A company manufactures two types of electronic components: Type A and Type B. Due to supply chain constraints, only 40% of the components produced are Type A, and the rest are Type B. The production process is such that 5% of Type A components and 10% of Type B components are defective.

The company conducts a quality control test where a component is randomly selected and tested for defects.

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- i. What is the probability that a randomly selected component is defective?
- ii. If a component is found to be defective, what is the probability that it is a Type A component?
- iii. Suppose the company implements a new quality improvement program that reduces the defect rate of Type B components to 6%, while the defect rate for Type A components remains the same. Recalculate the probability that a randomly selected component is defective.
- iv. With the new defect rates, if two components are randomly selected, what is the probability that at least one of them is defective?
- v. The company wants to ensure that the probability of selling a defective component is less than 4%. They can adjust the production ratio of Type A and Type B components while keeping the defect rates from part iii. What should be the new percentage of Type A components produced to achieve this goal?

End of the tutorial.